

Large Ultrathin Polyoxomolybdate-Decorated Boron Nitride Nanosheets with Enhanced Antibacterial Activity for Infection Control

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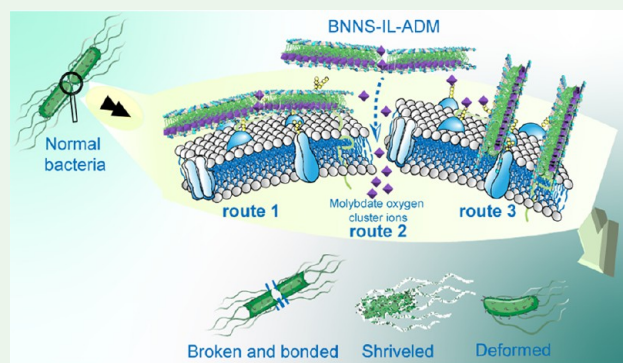
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ABSTRACT: To address the challenges in bacterial infection for killing germs without using antibiotics, the development of an efficient and safe antibacterial nanomaterial to control severe bacterial infection has become a critical issue. Herein, ionic liquid (IL)-promoted simultaneous exfoliation and conjugation of boron nitride nanosheets (BNNSs) with ammonium dimolybdate (ADM) are developed. Such an approach provides large ultrathin nanosheets with three to four layers and micron lateral dimensions and facilitates intercalation with a molybdenum (Mo) content of 6.25%. BNNS-IL-ADM provides nanoconfinement inducing capability for sustained release of Mo ions to decelerate the cumulative release rate by 30.45%. BNNS-IL-ADM imparts a superior antibacterial effect especially at a low concentration level of bioactive components, which improves the antibacterial efficiency by 92.00% against Gram-negative *Escherichia coli* and Gram-positive *Staphylococcus aureus* in comparison with ADM. The enhanced antibacterial performance is strongly dependent on the synergistic effect between the BN nanocarrier and bioactive ADM, in which the bacterial cell adhesion induced by the inherent hydrophobic property of the BNNS and the nanochannels within BNNS-IL-ADM protecting the bioactive ingredient and achieving a sustained release play significant roles. BNNS-IL-ADM can serve as a safe and efficient antibacterial agent with very low cytotoxicity. Such features offer a feasible strategy for simultaneous exfoliation and functionalization of boron nitride nanosheets as a promising platform for the delivery of bioactive ingredients against bacterial infections.

KEYWORDS: boron nitride nanosheets, nanoconfinement effect, polyoxometalates, sustained release, antibacterial mechanism



1. INTRODUCTION

Bacterial infection poses a great threat on the public health safety and ecological environment.^{1,2} Nevertheless, the widespread use of traditional antibiotics has caused an increase in bacterial resistance and also results in the emergence of superbugs, which make it extremely difficult to treat the infection.^{3–5} Therefore, it is desired to develop alternative strategies with high efficiency, environmental friendliness, and nontoxic side effects for the control of bacterial infection and combating antibiotic resistance crisis.

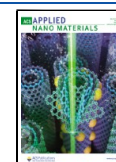
In response to this crisis, a tremendous amount of effort has been devoted to the design of antibacterial materials, including metals,^{6–8} peptides,⁹ biopolymers,^{10,11} engineered nanomaterials,^{12,13} etc. Polyoxometalates (POMs) are unique polyanionic clusters composed of early transition metal ions bridged by oxygen atoms, which are considered promising metallodrugs due to their wide range of tunable structures and biological effects.¹⁴ In particular, POMs have demonstrated surprising biological advantages in the treatment of infections associated

with bacteria, which exert antibacterial activity through penetration.^{15,16} Several POMs, especially polyoxomolybdates, have recognizable antibacterial activity against both Gram-negative and Gram-positive bacteria, depending on their composition, morphology, and size.^{17,18} Ammonium molybdate, a kind of polyoxometalate, can be regarded as a water-soluble metal oxide with an oxygen-rich surface, whose structure, composition, and charge properties can be adjusted.¹⁹ However, the excellent water solubility of POMs limits their processability and leads to leaching concerns during applications.

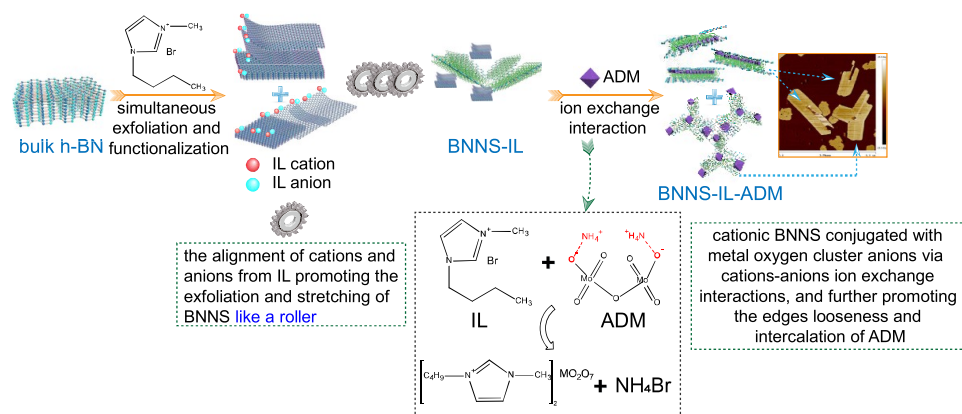
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Scheme 1. Schematic Illustration of the BNNS-IL-ADM Formation Mechanism and Ion Exchange Process



To address this issue, polyoxometalate-decorated inorganic materials have been in the spotlight. POMs anchored on the surface of solid supports or carriers are preferable to enhance the properties owing to the facile accessibility of substrates/substances to the active sites.^{20–22} For instance, Xing et al. designed copper-substituted polyoxomolybdate-decorated graphene oxide nanocomposites by a facile and green synthetic strategy, which showed the dual functionality of excellent antibacterial properties.²⁰ POMs are usually loaded into carriers by means of impregnation, adsorption, ion exchange, etc.²³ Particularly, chemical bonding, that is, premodifying the surface of the carrier channels with functionalized cations, can significantly increase the effective loading of POMs, improve the dispersion of POMs in the system, and thus inhibit POM leaching.^{24–26} In comparison with pure POMs, POM-decorated inorganic materials provide advantages with high stability to protect POMs from leaching or being degraded and also reduce toxicity toward normal cells.²⁷ However, controlled ion release within POM-decorated functional materials is not easy to achieve and affected by several factors, such as the bactericide element content, state of its agglomeration, surface topography, and roughness.^{28,29} Thus, it still remains a major challenge to develop POM-decorated functional materials with high efficacy against various pathogens and capabilities of initial fast bactericide release to rapidly inhibit infection and then long-term defense to prevent latent infection at a low total dosage.³⁰

Recently, nanomaterials with various nanostructures, such as nanoparticles, nanotubes, and nanosheets, have attracted great interest as promising drug carriers or antibacterial agents, attributed to their unique physicochemical, optical, electrical, and mechanical properties as well as tunable structures.^{31–37} Among them, hexagonal boron nitride (h-BN), as a kind of two-dimensional material, exhibits outstanding biocompatibility, environmental safety, and lower cytotoxicity in comparison with carbon-based nanomaterials.^{38–41} Despite high chemical inertness and strong oxidation resistance, it is still possible for BN nanosheets to biodegrade in the lungs.⁴² h-BN has a lamellar structure with a hexagonal honeycomb shape composed of boron (B) and nitrogen (N) atoms, which in combination with homogeneous interlayer channels and stacked layers, tunable surface modification, adjustable structures, and stable properties can serve as one of the promising inorganic nanocarriers.^{43–46} In our previous work, boron nitride nanosheets (BNNSs) exfoliated from h-BN possess abundant edge defect sites and hydrophobic proper-

ties, which facilitate the introduction of functional groups and the loading of active ingredients through hydrophobic and π – π interactions.^{47–49} Additionally, BNNSs with fewer layers by exfoliation possess considerable mechanical properties and sharp edges, which can cause physical damage in bacterial cells, also realize a synergistic bacteriostatic effect with active ingredients, and thus improve antibacterial activity.³⁰ Moreover, the lamellar properties of the BNNSs are favorable for spreading and adhesion on various surfaces, thus reducing the deterioration, degradation, and leaching off of active ingredients.⁴⁹ As reported, the biocompatibility and loading efficiency of BN nanosheets depend on their size, thickness, and surface chemical properties.^{50–53} For example, Pan et al. reported that BN nanosheets had antimicrobial resistance-independent antibacterial activity without triggering secondary resistance and displayed excellent biocompatibility in mammals.⁵³

However, the strong van der Waals forces between the pristine h-BN lamellae block the massive production of BN nanosheets and limit the application prospects as a nanocarrier for drug delivery.⁵⁴ Although the micron dimensional size of h-BN provides a large immobilization space for the cargo (i.e., POMs), it is difficult to control the loading and release of the cargo by only relying on the hydrophobic interaction and π – π interaction. Due to the lack of a strong interaction and bulk characteristic of pristine h-BN, the immobilized compounds typically show a low loading amount and a fast and uncontrolled release.⁵⁵ In general, the traditional exfoliation treatments are cumbersome processes and consume huge amounts of caustic solvents or additives.^{56,57} Ionic liquids (ILs) have the advantages of nonvolatility, solubility, and electrochemical stability, which are reported as candidate materials for environmentally friendly solvents to replace caustic solvents.⁵⁸ In addition, the cationic room temperature molten salt provides an action site for ion exchange due to the electronic holes. Hence, it is urgent to design and develop a facile and green strategy to exfoliate and functionalize pristine h-BN and then strengthen the interaction between the BN-based materials and cargo.

In this study, IL-promoted simultaneous exfoliation and functionalization of BNNSs (BNNS-IL) were realized via cation-mediated interactions between ILs and the partly ionic BN surface as well as affinity and force at the interface orientated from their similar surface energies. Then, BNNS-IL was conjugated with polyoxomolybdates (BNNS-IL-ADM) via cation–anion ion exchange interactions between metal oxygen

cluster anions and imidazolium cations, which further promoted the edge looseness and intercalation of ADM, leading to the formation of large ultrathin nanosheets with three to four layered nanosheets and micron lateral dimensions and acquiring a large aspect ratio of 325 (Scheme 1). The loading and nanoconfinement-induced sustained release of bioactive ingredients within BNNS-IL-ADM was analyzed. The antibacterial activity of BNNS-IL-ADM was systematically investigated against Gram-negative and Gram-positive bacterial strains, and the antibacterial mechanism was proposed on the basis of concrete experimental results.

2. EXPERIMENTAL SECTION

2.1. IL-Promoted Exfoliation and Functionalization of Boron Nitride Nanosheets. 10 g of IL was added into a three-necked flask and heated at 120 °C to be melted into a colorless and transparent viscous liquid. Then, 0.3 g of BNNS was added to the IL solution, and the mixture was subjected to magnetic stirring at 120 °C for 5 h. After cooling to room temperature, the obtained product was centrifuged at 6000 rpm for 15 min, the supernatant containing the ionic liquid was collected and recovered for recycling application, and then, the white precipitate was washed using deionized water three times and dried at 70 °C for 48 h for further removal of the remaining solvents to obtain ionic liquid-promoted simultaneously exfoliated and functionalized boron nitride nanosheets (BNNS-IL).

2.2. Preparation of BNNS-IL-ADM. The IL promotes the exfoliation and functionalization of BNNS due to its similar surface energy to h-BN; also, the IL can interfere the van der Waals force between the boron nitride layers and bind to the B and N atoms with vacancies as a bridge for ion exchange with polyoxometalates.

In a typical procedure, first, 0.5 g of ADM was dispersed in 20 mL of deionized water and stirred to obtain a colorless and transparent stock solution. Then, 0.15 g of the as-obtained BNNS-IL aqueous dispersion by ultrasonication was slowly added into the ADM stock solution, and subsequently, ion exchange was conducted initially at 30 °C with vigorous magnetic stirring for 1 h. Afterward, the mixture solution was transferred to a Teflon-lined stainless-steel autoclave and reacted at 80 °C for 24 h to further conduct the ion exchange reaction. After completion and cooling to room temperature, the product was centrifuged at 6000 rpm for 15 min to remove the supernatant. Then, the precipitate was collected and washed with ethanol three times and dried by freezing drying to acquire the ultimate BNNS-IL anchored and intercalated with molybdate oxygen cluster ions of ADM (BNNS-IL-ADM or BNNS-IL-ADM-0.50). The effect of ADM addition was also investigated by altering the initial ADM additive amount to 0.25 and 1.00 g; the other procedures were still repeated as mentioned above, and the products were marked as BNNS-IL-ADM-0.25 and BNNS-IL-ADM-1.00, respectively.

2.3. Characterizations. The morphology and dimensions were observed using scanning electron microscopy (SEM) (EVO 18, ZEISS) after spraying gold for 45 s and transmission electron microscopy (TEM) (TF20 2100F, JEOL). Energy dispersive spectroscopy (EDS) analysis was performed on an Oxford Instruments (X-MAXn65 T). The surface morphology, phase, layers and thickness of BNNS-IL-ADM and BNNS-IL were examined by atomic force microscopy (AFM) (Bruker Dimension Edge) in tapping mode. ζ -potential measurements were performed by dispersing and diluting samples in Milli-Q water or ethanol and carried out via a 90 Plus PALS particle size and ζ -potential analyzer (Bruker Corporation) at a scattering angle of 90° at 25 °C. The crystallinity and the d -spacing were analyzed using Cu K α X-ray diffraction (XRD) (D8 diffractometer, Bruker AXS GmbH) with 2θ between 5 and 80°, and the d -spacing was calculated based on Bragg's law, $2d \sin \theta = n\lambda$. The chemical interactions were determined by Fourier transform infrared (FTIR) (Spectrum 100, PerkinElmer Inc.) with the KBr pellet method to compare the vibrational states with a resolution of 4 cm⁻¹ in the range of 4000–400 cm⁻¹. The structural properties mainly including the G and D bands were characterized by Raman

spectroscopy (DXR2, Thermo Fisher) in a detector with a 532 nm laser for excitation. X-ray photoelectron spectroscopy (XPS) was carried out with a K-Alpha spectrometer (Thermo Scientific) under vacuum using monochromatic Al K α X-rays to obtain the information about chemical composition and bonding chemistry for the surface layers. Ultraviolet–visible diffuse reflectance spectroscopy (UV–vis DRS) measurements were carried out using a UV-3600 spectrophotometer (Shimadzu) equipped with an integrating sphere attachment BaSO₄ as a reflectance standard within 200–800 nm. Thermogravimetric analysis (TGA) was performed on a TGA2 thermogravimetric analyzer (Mettler Toledo) under a N₂ atmosphere with a flow rate of 20 mL·min⁻¹, heating from 40 to 600 °C at a heating rate of 10 °C·min⁻¹ to compare the thermal stability and compositions. DSC was carried out using a Q20 differential scanning calorimeter (TA Instruments) by heating from 40 to 270 °C at a heating rate of 10 °C·min⁻¹ under a nitrogen atmosphere of 50 mL·min⁻¹. The content of Mo was also determined by inductively coupled plasma optical emission spectroscopy (ICP-OES) (Agilent 720ES), where BNNS-IL-ADM was dissolved in 10 mL of aqua regia (HNO₃/HCl = 1/3 v/v) for 5 min, and then, the solution was diluted and filtered.

2.4. Study of Antibacterial Activity. The antibacterial activity of free ADM and BNNS-IL-ADM with an equivalent content of Mo was determined by the agar plate diffusion method.^{59,60} Gram-negative *Escherichia coli* and Gram-positive *Staphylococcus aureus* (0.1 mL) were inoculated in Luria Bertani (LB) broth with an initial cell density of 1×10^6 CFU·mL⁻¹ and incubated at 37 °C for 24 h. Then, *E. coli* and *S. aureus* were evenly spread on agar plates. Afterward, 20 mg of BNNS-IL-ADM was placed into a 6 mm-diameter hole in the center of the agar plates, and then, all Petri dishes for testing were incubated at 37 °C for 24 h. The diameter of the inhibition zone was measured. As a control, equivalent free ADM was calculated in terms of loading by ICP-OES analysis.

In addition, the same amounts of h-BN, BNNS, BNNS-IL, and IL were also used to compare the inhibition activity of *E. coli* and *S. aureus*. The inhibition of bacteria with samples was evaluated by measuring the diameter of the bacteriostatic ring, from which it was observed that the samples exhibited better antibacterial activity if the inhibitory region was larger. All antibacterial tests in this study were conducted in triplicate.

2.5. Study on the Minimal Inhibitory Concentration (MIC) against Bacteria. Antibacterial activity of BNNS-IL-ADM was quantitatively analyzed by determining the minimal inhibitory concentration (MIC) in accordance with a previously reported method.^{61,62} In a typical process, a certain amount of BNNS-IL-ADM powder was dispersed into 20 mL of sterilized nutrient agar under ultrasonication at 45 °C for 10 min to obtain dispersions with various concentrations in the range of 0.08–0.24 and 0.20–0.36 mg·mL⁻¹. Then, the dispersions were poured into 0.1 mL of *E. coli* and *S. aureus* suspension at a concentration of 1×10^6 CFU·mL⁻¹ after agar plate coagulation. Next, the mixture was incubated at 37 °C for 24 h in the dark. Then, the numbers of colonies in the culture dishes were observed, and the corresponding MIC value was determined by measuring the minimal concentration of the sample solution at which the cells had no bacterial growth of *E. coli* and *S. aureus*. The antibacterial rate (%) was calculated from the inhibition number of *E. coli* and *S. aureus* cell colonies by BNNS-IL-ADM and free ADM through comparison of the blank without any ADM samples, which was calculated according to eq 1, where C_{blank} represents the number of bacterial cell colonies without any ADM samples after incubation for 24 h and C_A represents the number of bacterial cell colonies after 24 h of incubation with BNNS-IL-ADM and free ADM, respectively.

$$\text{antibacterial rate (\%)} = \frac{C_{\text{blank}} - C_A}{C_{\text{blank}}} \times 100\% \quad (1)$$

2.6. Evaluation of the Growth Curve of Bacteria. The antibacterial activity of free ADM and BNNS-IL-ADM against *E. coli* was also characterized by the growth inhibition curve method.^{63–65} After the LB broth was sterilized, 0.1 mL of *E. coli* (1×10^6 CFU·mL⁻¹) was inoculated into the blank control (only LB broth). Then,

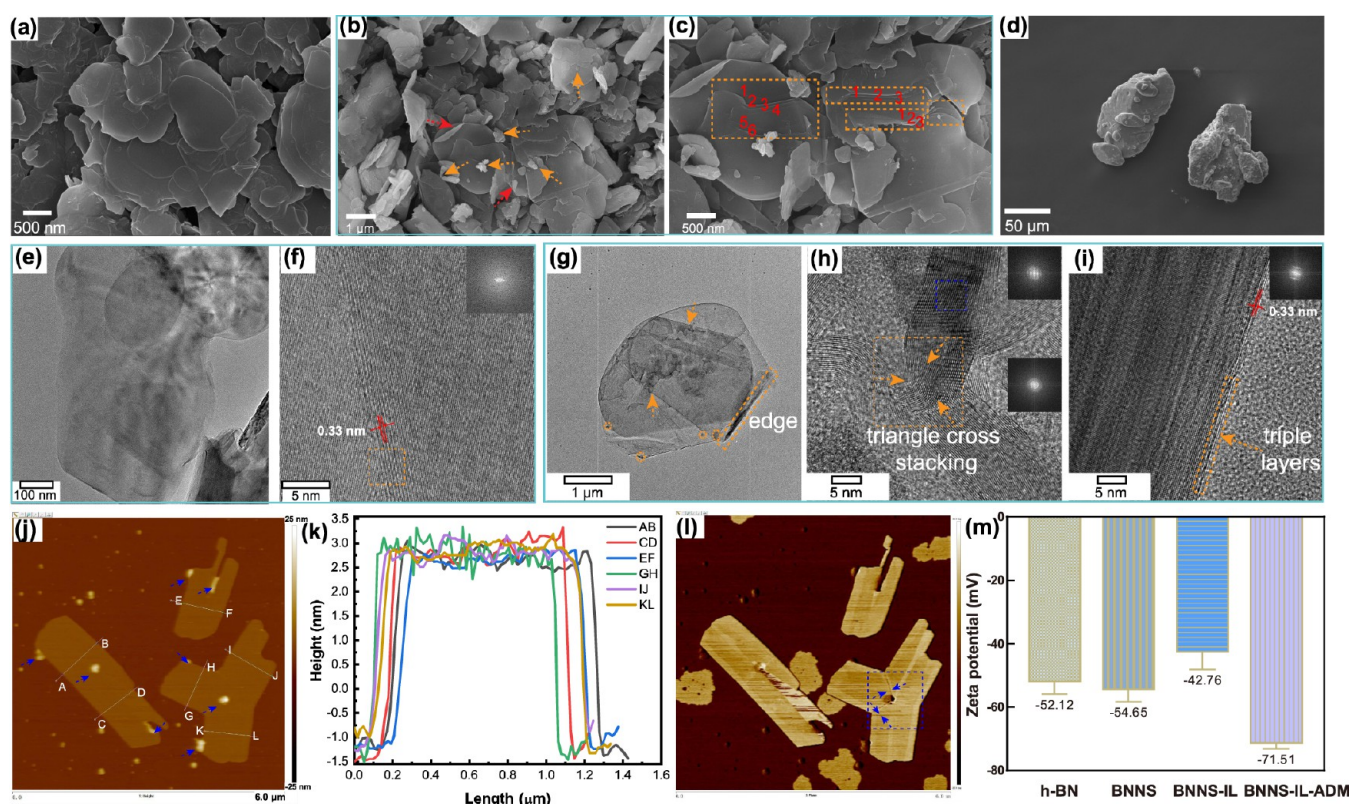


Figure 1. SEM images of (a) BNNS-IL, (b, c) BNNS-IL-ADM, and (d) ADM; TEM and HR-TEM images of (e, f) BNNS-IL and (g–i) BNNS-IL-ADM under different magnifications; the insets in (f), (h), and (i) are their corresponding fast Fourier transform (FFT) patterns; (j–l) AFM micrograph, topographical height profiles, and phase image of BNNS-IL-ADM; and (m) ζ potentials of h-BN, BNNS, BNNS-IL, and BNNS-IL-ADM dispersed in an ethanol anhydrous solution.

certain amounts of ADM and BNNS-IL-ADM with an equivalent Mo content as ADM were added into 3 mL of LB broth and incubated in a shaker at 37 °C for 24 h at a speed of 150 rpm. The optical density (OD) value at a wavelength of 600 nm was recorded using a microplate reader at certain time intervals at 37 °C for 24 h. Each point was repeated in triplicate.

2.7. Antibacterial Mechanism Evaluation. **2.7.1. Monitoring the Morphological Changes in Bacterial Cells.** The morphological changes of Gram-negative bacterial (*E. coli* as a model) cells treated with BNNS-IL-ADM were monitored using biological SEM (Sigma 300, ZEISS). Specifically, BNNS-IL-ADM powder with a concentration of 1.5 times MIC was added to the unsolidified LB agar medium, which was dispersed uniformly by ultrasonication and poured into a Petri dish for solidification. Afterward, 0.1 mL of *E. coli* suspension with a concentration of 1×10^6 CFU·mL⁻¹ was slowly added into the culture dish and inoculated on the surface of LB agar and then cultured at 37 °C for 6 h. After completion, the agar containing BNNS-IL-ADM and *E. coli* was fixed with 2.5% glutaraldehyde solution overnight at 8 °C. After dehydrating the samples with gradient concentrations of 30, 50, 70, 80, 90, and 95% ethanol for 15 min, the samples were dried overnight and gold was sputtered onto the sample surfaces. The interaction between the samples and *E. coli* was observed by SEM, and the morphology of *E. coli* without any treatment was used as a control.

2.7.2. Bacterial Cell Staining Assay. To further explore the antibacterial mechanism of BNNS-IL-ADM against Gram-negative bacteria (*E. coli* as a model), the culture medium of *E. coli* treated with BNNS-IL-ADM with a concentration of MIC values was stained with dyes, propidium iodide (PI) and SYTO-9, for 15 min in a dark environment, and the images were examined and recorded using a confocal fluorescence microscope (FV1200, Olympus). Basically, for bacterial stain observation, the concentration of the *E. coli* suspension was diluted to 1×10^6 CFU·mL⁻¹ using phosphate-buffered saline (PBS) and mixed with ADM and BNNS-IL-ADM dispersions for 24 h

at 37 °C. After culturing, the culture solution was centrifuged at 8000 rpm for 3 min, and the supernatant was discarded. Then, 300 μL of PI and SYTO-9 was added and mixed evenly with staining at 37 °C for 15 min. The dead bacteria were stained with PI, showing red fluorescence at an excitation wavelength of 599 nm and emission wavelength of 620 nm. The living bacteria were stained with SYTO-9, revealing green fluorescence at an excitation wavelength of 488 nm and emission wavelength of 503 nm.¹⁵ In addition, *E. coli* treated with 1% dimethyl sulfoxide (DMSO) solution was used as a blank control.

2.8. In Vitro Cytotoxicity Assay. The cytotoxicity effect of BNNS-IL-ADM toward NIH3T3 was evaluated in vitro using the colorimetric 3-[4,5-dimethylthiazol-2-yl]-2,5-diphenyltetrazolium bromide (MTT) assay. NIH3T3 cells were seeded on 96-well plates at an initial density of 5×10^3 cells per well in Dulbecco's modified Eagle medium (DMEM, 5 mL) with 10% fetal bovine serum (FBS) and incubated at 37 °C in 5% CO₂ for 8 h in a humidified incubator until the cells were full in the culture incubator. Afterward, the culture medium was removed and replaced with fresh DMEM medium (5 mL) containing different concentrations of BNNS-IL-ADM in the range of 50–750 mg·L⁻¹, and the cells were further continuously cultured at 37 °C in 5% CO₂ for 24 h. After 24 h, the cells were rinsed twice with saline (PBS), and then, MTT solution (200 μL, 0.5 mg·mL⁻¹) was added to each well and incubated for an additional 4 h. Then, the medium was removed and dimethyl sulfoxide (DMSO, 150 μL) was added to solubilize the formazan salt. Untreated cells were used as a control. The absorbance was measured at a wavelength of 570 nm using a microplate reader. Relative cell viability (%) was determined by comparing the absorbance values between treated cells and control cells. Cell viability was calculated using eq 2, where OD_{sample} and OD_{control} refer to the optical density of treated cells with BNNS-IL-ADM and untreated control cells, respectively. Each experiment was repeated three times.

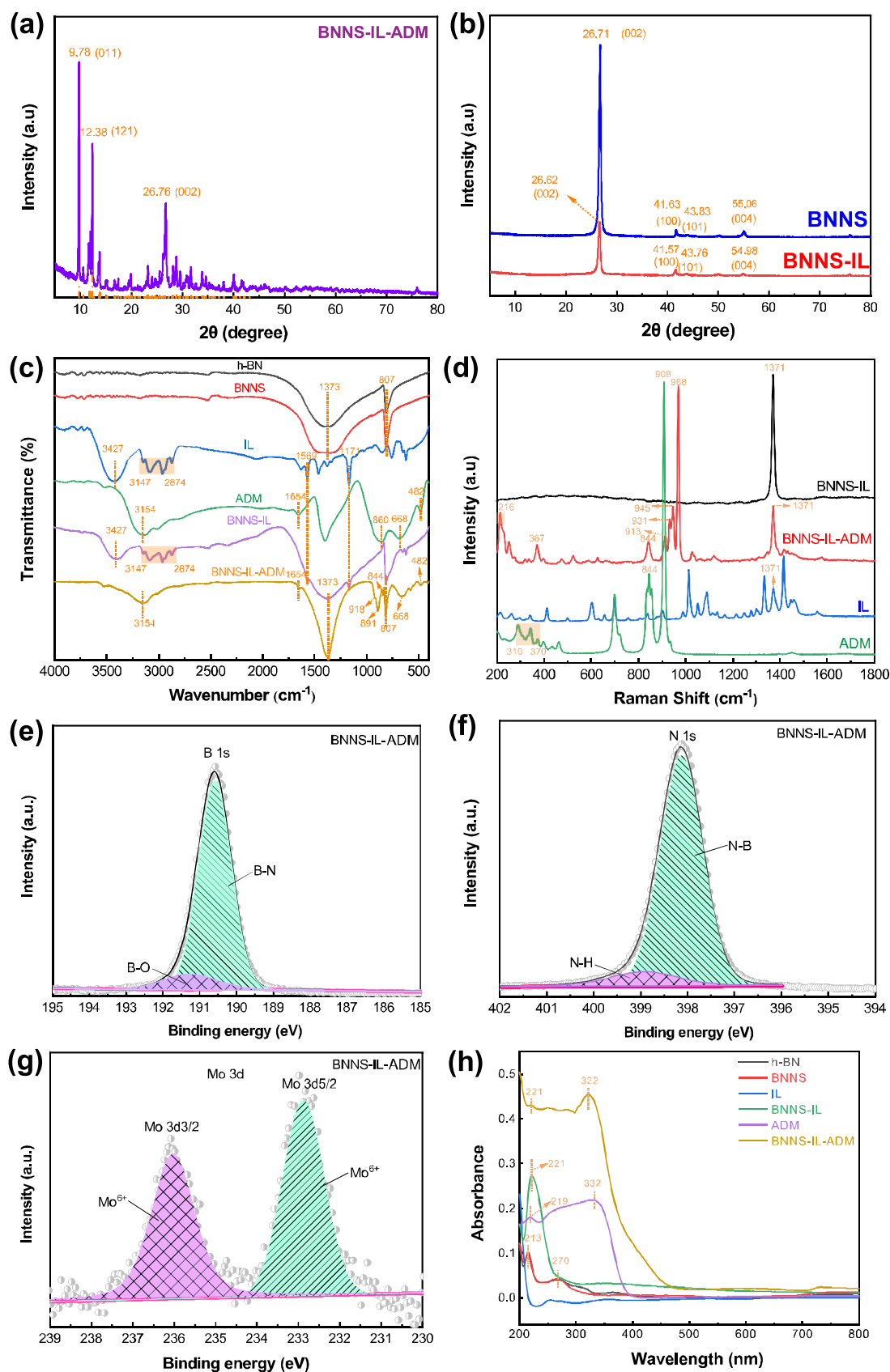


Figure 2. XRD patterns of (a) BNNS-IL-ADM and (b) BNNS and BNNS-IL, (c) FTIR spectra of virgin h-BN, BNNS, pure IL, pure ADM, BNNS-IL, and BNNS-IL-ADM, (d) Raman spectra of pure IL, pure ADM, BNNS-IL, and BNNS-IL-ADM, high-resolution XPS spectra of (e) B 1s, (f) N 1s, and (g) Mo 3d electrons for BNNS-IL-ADM, and (h) UV-vis diffuse reflectance spectra of virgin h-BN, BNNS, pure IL, pure ADM, BNNS-IL, and BNNS-IL-ADM.

$$\text{cell viability (\%)} = \frac{\text{OD}_{\text{sample}}}{\text{OD}_{\text{control}}} \times 100\% \quad (2)$$

3. RESULTS AND DISCUSSION

3.1. Surface Morphology and Nanostructure. This approach via IL-promoted simultaneous exfoliation and conjugation with metallic ions, i.e., ammonium dimolybdate (ADM), can effectively achieve the few-layered BNNS with micron lateral dimensions, intuitively observed from scanning electron microscopy (SEM), transmission electron microscopy (TEM), high-resolution TEM (HR-TEM), and atomic force microscopy (AFM) images by comparing in Figures 1 and S1–S3. Unlike strong sonication, damaging the BN framework and leading to small lateral size, BNNS-IL via IL-assisted exfoliation shows partially wrinkled and sharper edges and rougher and disordered surfaces with larger lateral size as well as more transparent surfaces with approximately six BNNS layers (Figure 1a,e,f). In contrast, virgin h-BN and BNNS only via ultrasonication exhibit smooth and relatively regular edges and flat surfaces with a tightly stacked structure (Figure S1a,b,c–f). Such observation reveals that the IL-assisted exfoliation strategy facilitates the movement of stacked layers via cation-mediated interactions between ILs and the partly ionic BN surface as well as affinity and force at the interface orientated from their similar surface energies, favoring the ordered alignment of cations and anions within BN nanochannels via nanoconfinement, crippling the Coulomb force between the cations and anions of the IL.^{58,66} Nevertheless, some nanosized polyhedrons and irregular fragments attach onto the surface once BNNS-IL conjugated with ADM, highlighted with orange and red arrows (Figure 1b), in complete contrast to the micromorphology of bulk polyhedron aggregates (Figure 1d), indicative of planar BNNS-IL, providing enough and continuous space for ADM intercalation and inhibiting their aggregation via weakening their attraction force. Meanwhile, the attachment and stacking of ADM mainly occur at the edges of BNNS, ascribed to the active edge defect sites and vacancies via cation–anion interactions between metal oxygen cluster anions and imidazolium cations, and further promote the edge looseness and intercalation of ADM (indicated by red arrows in Figure 1b), leading to the formation of counted terraced few-layered nanosheets (Figure 1c). The corresponding energy dispersive spectroscopy (EDS) analysis shows the uniform distribution of the Mo element along with N and O elements in Figure S2, indicative of the uniform decoration and intercalation of metal oxygen cluster ions within BNNS-IL nanosheets.

Furthermore, the distribution of ADM clusters can be identified from the darker dots appearing at the edges and on the transparent surface of BNNS-IL (circled and arrowed in orange) from the TEM image in Figure 1g. The introduction of ADM also leads to the wrinkled edges (marked by the orange dashed box). Intriguingly, the triangle cross-stacking lattices are generated, rooting in the interfaces of the overlapping region of three layers for BNNS-IL-ADM, ascribed to $[\text{Mo}_2\text{O}_7]^{2-}$ clusters anchored along the dimensions of BN nanochannels and connecting their edge interfaces. This reason leads to lattice fringe whirling and intertwining and results in the fast Fourier transform (FFT) pattern exhibiting sharp spotted concentric diffraction rings with polycrystalline structures (Figure 1h). Instead, the central areas and most of the edge areas maintain relatively regular interatomic spacing

and lattice fringes with a similar lattice distance of ~ 0.33 nm (Figure 1h,i) in comparison with h-BN, BNNS, and BNNS-IL observed and estimated from HR-TEM micrographs (Figures 1f and S1d,f).

Likewise, from the phase image of AFM, the assembled edges and binding corner sites stemming from ADM clusters highlighted with dark-blue arrows (Figure 1l) can be observed, interpreting the formation of the triangle cross-stacking region in HR-TEM. The thickness of BNNS-IL-ADM is estimated to range from ~ 3 to 4 nm by a height profile of AFM scanning (Figure 1j,k), corresponding to three to four layered nanostructures, considering that the thickness of a single BN monolayer is $0.9\text{--}1.0$ nm,^{67,68} which is similar with the HR-TEM image revealing triple counted layers in Figure 1i. Instead, the thickness of BNNS-IL in Figure S3 is approximately 1.70 ± 0.02 nm. The thickness change tendency is presumably due to the decoration and intercalation of ADM within BNNS-IL nanochannels. Additionally, as estimated from Figure 1j,k, the average lateral dimension of BNNS-IL-ADM is determined to be $\sim 925 \pm 50$ nm, in agreement with the case of BNNS-IL in Figure S3 with the value of $\sim 929 \pm 82$ nm. As evaluated, BNNS-IL-ADM still experiences a high aspect ratio of $\sim 325 \pm 50$.

3.2. Colloidal Stability. The ion exchange interaction appearing between BNNS, IL, and ADM can be preliminarily interpreted by ζ -potential measurements (Figure 1m). Virgin h-BN exhibits abundant negative charges with a value of -52.12 ± 3.85 mV, due to the ionic character between B–N bonds. The BNNS under liquid ultrasonication introduces some –OH groups with more negative charges (-54.65 ± 3.70 mV). The imidazolium ring cations stemming from ILs attract partially negative charges of the BNNS via electrostatic complexation and π – π stacking, inducing reduction in the absolute magnitude to -42.76 ± 5.50 mV. The introduction of the imidazolium cations onto the BNNS provides more probabilities to bind with ADM via ion exchanges, evidenced from the magnification in the absolute magnitude to -71.51 ± 1.72 mV, which may increase charge transport across the interface. The much larger negative charges provide steric hindrance against aggregation and manifest better colloidal dispersibility. The colloidal dispersibility of BNNS-IL and BNNS-IL-ADM in deionized water and bacterial culture medium (10% broth as model medium) is evidenced from photographs and Tyndall scattering analysis (Figures S4a–d and S5). BNNS-IL-ADM both in 10% broth and water results in a stronger light path from Tyndall scattering in comparison with BNNS-IL, confirming that BNNS-IL-ADM exists in a steady colloid state within 3 days.

3.3. Chemical Interactions and Compositions. To further confirm the crystalline and phase microstructural evolutions, wide-angle X-ray diffraction (XRD) was performed and XRD patterns are shown in Figures 2a,b and S6a. BNNS-IL still exhibits a typical hexagonal structure with a peak at 26.62° and a lattice distance of ~ 0.33 nm for the (002) plane, evidenced from similar XRD patterns to those of h-BN and BNNS in Figures S6a and 2b, suggesting that IL-assisted exfoliation can retain the BN ordered crystalline structure. In addition, a wider full width at half maximum (FWHM) for BNNS-IL is observed at 0.435 versus the case of pristine h-BN with the value of 0.344. Nevertheless, the peak at 26.66° corresponding to the (002) plane stemming from the virgin h-BN crystal shifts to 26.76° with the unchanged lattice spacing for BNNS-IL-ADM, a sharp peak with significantly reduced

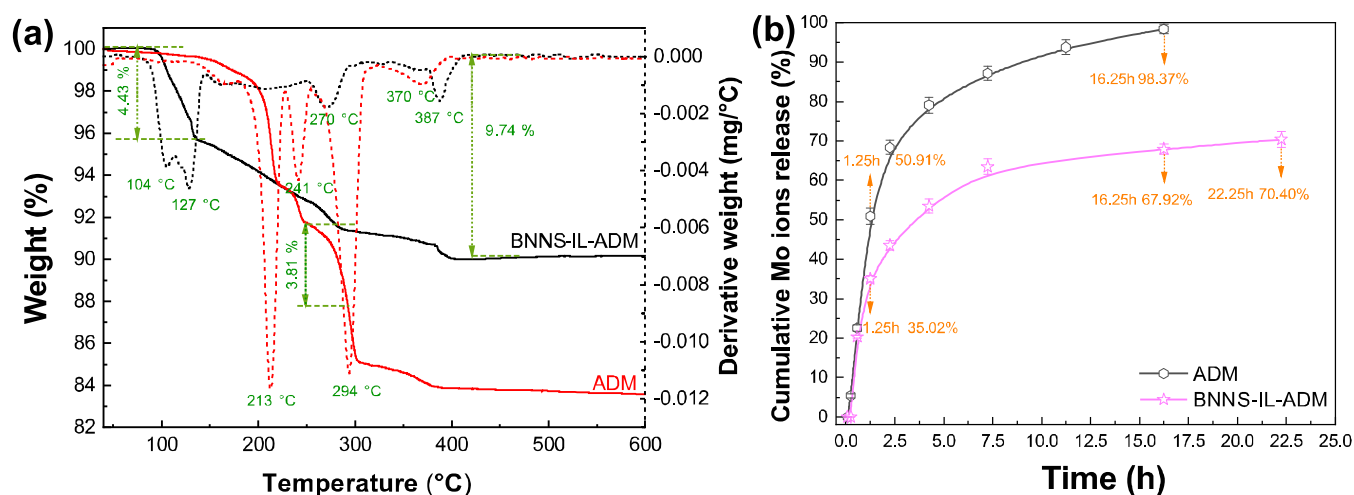


Figure 3. (a) TG (solid line) and DTG (dashed line) curves of pure ADM and BNNS-IL-ADM and (b) cumulative Mo ion release curves from free ADM and ADM-loaded BNNS-IL (BNNS-IL-ADM) in soil column experiments.

crystalline intensity (Figure 2a). A slightly narrower FWHM for this peak is observed to be 0.421 by comparing with BNNS-IL. Additionally, in Figure 2a, another sharp and strongest characteristic reflection peak at 9.78° ($d = 0.90$ nm) is generated corresponding to the (011) plane originating from ADM with a FWHM of 0.171,⁶⁹ which suggests the successful nanoconfinement and intercalation of ADM clusters within the BN nanochannels holding their identical crystalline structures, leading to the formation of polycrystalline structures.⁷⁰

Fourier transform infrared (FTIR) spectroscopy was carried out to investigate the chemical interactions between BNNS, IL, and ADM by comparing the evolution of chemical functional groups, as depicted in Figure 2c. The FTIR spectrum of BNNS-IL preserves the intrinsic characteristic peaks of in the plane B–N stretching vibrations at nearly 1373 cm^{-1} and out-of-plane B–N–B bending vibrations at about 807 cm^{-1} . The emergence of new characteristic peaks, including a new absorption band at 3427 cm^{-1} , a series of bands located between 3147 and 2874 cm^{-1} , and other absorption bands at 1569 and 1171 cm^{-1} within BNNS-IL, is ascribed to the introduction of imidazolium cations formed by the hydrogen bonding between the empty orbitals provided by nitrogen atoms and halogen anions and the asymmetric and symmetric stretching vibrations of aliphatic groups from ILs, along with the stretching vibrations of C=C and C=N groups from the imidazolium ring. Such changes are indicative of BNNS confining ILs via cation–BN interactions, hydrogen bonds, and π – π stacking interactions between ILs and BN nanochannel surfaces. Likewise, the FTIR spectrum of BNNS-IL-ADM still holds the strong BN intrinsic characteristic peaks centered at 1373 and 807 cm^{-1} , whereas the former represents evidently sharper and narrower characteristics in comparison with the peaks of h-BN, BNNS, and BNNS-IL, which is attributed to the anchoring of ADM within BN nanochannels via electric interactions. An additional change appears at 3154 cm^{-1} assigned to the N–H stretching vibration (NH_4^+ in ADM), accompanied with the absence of the band at 3427 cm^{-1} , proving the existence of the ion exchange interaction between imidazolium cations and molybdate oxygen cluster anions.³⁸ In comparison with pure ADM, the reserved characteristic peaks at 668 and 482 cm^{-1} are ascribed to Mo–O–Mo for the surrounding molybdenum tetrahedron and central tetrahedron Mo–O–Mo bending vibration within $[\text{Mo}_2\text{O}_7]^{2-}$, respec-

tively.⁶⁹ Meanwhile, the bands splitting at 918 , 891 , and 844 cm^{-1} from one peak at 860 cm^{-1} represent the oscillation of terminal Mo=O stretching vibrations via multiple interactions within BNNS, IL, and ADM.

Raman spectra were acquired to gain further insights into the chemical interactions within BNNS, IL, and ADM, in Figure 2d. Compared to virgin h-BN and BNNS (Figure S6b), the G band at 1364 cm^{-1} ascribed to out-of-plane B–N vibrations (E_{2g}) is substantially upshifted to 1371 cm^{-1} for BNNS-IL and BNNS-IL-ADM systems, attributed to the exfoliation of h-BN into thin layers, leading to a higher in-plane strain and weaker interlayer interaction. As for bare ADM, the characteristic band is located at 908 cm^{-1} with a strong intensity assigned to Mo=O vibrations and 844 cm^{-1} assigned to Mo–O–Mo stretching modes and other bands associated with bending vibrations of terminal Mo=O groups and bridging Mo–O deformation vibrations. After ADM anchored onto BNNS-IL, the band located at 908 cm^{-1} shifts to 968 cm^{-1} shouldered with three splitting peaks at 945 , 931 , and 913 cm^{-1} , indicating a strong binding force via electrostatic and electrovalent interactions between molybdate oxygen cluster anions and BNNS-IL.

X-ray photoelectron spectroscopy (XPS) was applied to elucidate the surface chemical composition and element chemical state of BNNS-IL-ADM by comparing with h-BN and BNNS-IL. The XPS survey spectrum of BNNS-IL-ADM is displayed in Figure S7a, implying that it contains 48.47% of B, 39.17% of N, 12.04% of C, and 0.32% of Mo, whereas BNNS-IL involves 49.38% of B, 40.64% of N, and 9.98% of C (Figure S7b), which proved the successfully anchoring of ADM on the surface of the BNNS. Compared with virgin h-BN displaying a binding energy (BE) peak at 190.6 eV associated with B 1s and at 398.1 eV associated with N 1s,⁴⁹ the deconvolutions of B 1s and N 1s regions for BNNS-IL-ADM still exhibit the same BE peaks with the strongest intensities corresponding to B–N bonds, additionally with some B electrons and N electrons shifting to 191.3 and 398.9 eV , respectively, attributed to B–O between electron-deficient B atoms and Mo=O as well as N–H between BN and IL cations (Figure 2e,f). The deconvolution of the Mo 3d electron for BNNS-IL-ADM is presented in Figure 2g, which exhibits a spin–orbit doublet for the Mo valance state in the region of $3d_{5/2}$ and $3d_{3/2}$. The strong deconvolution peaks of Mo 3d located at 232.8 and

236.1 eV are ascribed to Mo^{6+} sites in the form of $[\text{Mo}_2\text{O}_7]^{2-}$, signifying Mo–O linkages at their edges.

Ultraviolet–visible diffuse reflectance spectroscopy (UV–vis DRS) was performed to further analyze the interactions within BNNS-IL-ADM, as presented in Figure 2h. BNNS-IL exhibits the intrinsic absorption band of h-BN and BNNS from 213 to 221 nm due to steric hindrance or the reflection of phonon–electron coupling, in the absence of the other peak onset at about 270 nm stemming from the structural or chemical defects in h-BN,^{49,71} signifying that the IL contributed to the oxidized regions of h-BN or boron vacancies. Instead, BNNS-IL-ADM presents the main absorption band centered at 322 nm shifting from 332 nm with respect to inherent $p\pi \rightarrow d\pi$ electron transition on the π -orbital of the $\text{Mo}=\text{O}$ bond at the end of ADM and the ligand–metal charge transfer, $\text{O}^{2-} \rightarrow \text{Mo}^{6+}$, implying that more photoinduced electrons are generated from UV–visible light illumination.⁷² Additionally, BNNS-IL-ADM still holds the intrinsic absorption peak of BNNS-IL at 221 nm with weaker intensity, suggesting the intercalation of Mo(VI) onto BNNS affecting the $\pi \rightarrow \pi^*$ electron transition.

3.4. Thermal Behavior. Thermogravimetric analysis (TGA) was performed under a N_2 atmosphere, and the corresponding derivative thermogravimetry (DTG) analysis curves were applied to investigate thermal stability and composition within BNNS-IL-ADM, as displayed in Figure 3a. The total amount of the IL within BNNS-IL is determined to be 12.80% by comparing the weight loss between h-BN, BNNS, and IL (Figure S8a–d), accompanied with the reduction of decomposition temperature from 304.0 to 270.9 °C, which is likely due to the nanoconfinement of the IL within BN nanochannels, decaying the Coulomb force between the cation and anion.⁶⁶ The TGA and DTG curves of pristine ADM in Figure 3a elucidate that the process of decomposition on heat treatment contains three strong degradation peaks centered at 213, 241, and 294 °C owing to ammonium cation decomposition and two weaker peaks at 164 and 370 °C, due to partial ammonium cations and molybdate anions, respectively, with a total weight loss of 16.43% at 600 °C and the residue existing in the MoO_3 state. However, upon forming BNNS-IL-ADM, the ADM amount can be determined to be $8.91 \pm 1.17\%$ within BNNS-IL-ADM by calculating the weight loss in the temperature range of 101–421 °C, attributed to the anchoring of partial ADM into the BNNS via ion exchange between the IL and ADM. In addition, the ion exchange process within BNNS-IL-ADM leads to the peak shifts and also shows two new peaks centered at 104 and 127 °C in the range of 83–145 °C. Such a change may be attributed to the introduction of $[\text{C}_8\text{H}_{15}\text{N}_2]^{2+}$ and NH_4^+ . Additionally, the Mo element amount within BNNS-IL-ADM measured by inductively coupled plasma optical emission spectroscopy (ICP-OES) is calculated to be $6.25 \pm 0.30\%$, a bit lower than the value obtained from the TG curve, which is likely to be a different treatment during the measurement (i.e., dissolution using aqua regia or thermal decomposition).

Differential scanning calorimetry (DSC) thermograms also support the TG and DTG results, as illustrated in Figure S8d. The melting point of the IL is 76.5 °C, while the IL promoted BN exfoliation (BNNS-IL), and the peak becomes very weak with a higher value of ~ 100 °C, owing to the extraordinary thermal stability of BN with the IL amount of 12.8%. In contrast to ADM, the DSC curve of BNNS-IL-ADM exhibits an upshift from 238.8 to 253.7 °C and downshift from 204.2 to

125.2 °C, where the former was attributed to the formation of new ammonium cations and the latter corresponded to the enhancement of thermal stability for the formation of MoO_3 .

As a green and stable molten salt, ILs have been reported to possess the potential to replace caustic solvents.⁷³ Also, for the sake of economic reasons, the recyclability of ILs for exfoliation and functionalization of BNNSs was investigated. Based on the quality difference of the dispersion before and after ultrasonication, the recovery rate of ILs after exfoliation of BNNSs for three cycles is 96, 95, and 91%. Figure S8c depicts the TG and DTG curves of BNNS-IL-ADM using the recovered IL for simultaneous exfoliation and ion exchange with ADM. The total weight loss percentage of BNNS-IL-ADM is 8.17, 5.12, and 7.40% in the temperature range of 40–600 °C using the IL for three cycles, which is only a little lower than the value of BNNS-IL-ADM using a fresh IL, indicating the good recycling utilization of ILs.

3.5. Nanoconfinement-Induced Sustained Release Behavior. The soil column leaching experiment imitating a confined environment was conducted via applying equivalent Mo ions in the form of free ADM and BNNS-IL-ADM, separately. The cumulative release curves of Mo ions under deionized water leaching through the soil column are depicted in Figure 3b. In comparison at an initial stage, the cumulative release rates of ADM and BNNS-IL-ADM are approximately similar owing to the existence of partial Mo ions on the surface of BNNS via weak interactions. Afterward, the cumulative release content of Mo ions dramatically increases with the increase of leaching time for the case of free ADM, which is close to complete release at 16.25 h. The rapid dissolution of ADM during the leaching process is attributed to its excellent water solubility. Instead, the nanochannels within BNNS-IL-ADM provide nanoconfinement inducing capability for sustained release of Mo ions. The release rate of Mo ions within BNNS-IL-ADM gradually reduces after 7.5 h in comparison with that of free ADM, i.e., only releasing 67.92% at 16.25 h and decelerating near 30.45%. The fundamental ion release kinetics is listed in Table S1, from which it can be concluded that the release of Mo ions from both free ADM and BNNS-IL-ADM is fitted well to the first-order kinetic model with the highest *R*-square among four models, indicating that the release rate is concentration-dependent. The time of half-release extends to 3.67 h for the case of BNNS-IL-ADM, whereas it is only 1.37 h for free ADM. The phenomenon is attributed to the nanoconfinement effect protecting active ingredients against leaching off as well as the strong adhesion ability between BNNS-IL-ADM and soil restricting the dissolution of Mo ions. The latter reason is proved by the appearance of soil with BNNS-IL-ADM in Figure S4e, where CK soil presenting a natural color is obtained from the soil column after the leaching experiment for free ADM, while the other one exhibits adhesion of white powder on the surface of soil. Furthermore, the release percentage of Mo from BNNS-IL-ADM in aqueous medium at 37 °C within 24 h is only $37.88 \pm 0.24\%$, exhibiting a sustained release behavior, Figure S9. Instead, free ADM is completely dissolved and released within 6 h through the dialysis membrane under aqueous conditions.

The various solvents (i.e., water, ethanol, ethanol–water mixture) were applied to redisperse the collected soil samples with BNNS-IL-ADM via ultrasonication (Figure S4f), and these soil suspensions were settled down for 10 h to show the sedimentation condition (Figure S4g), from which it can be

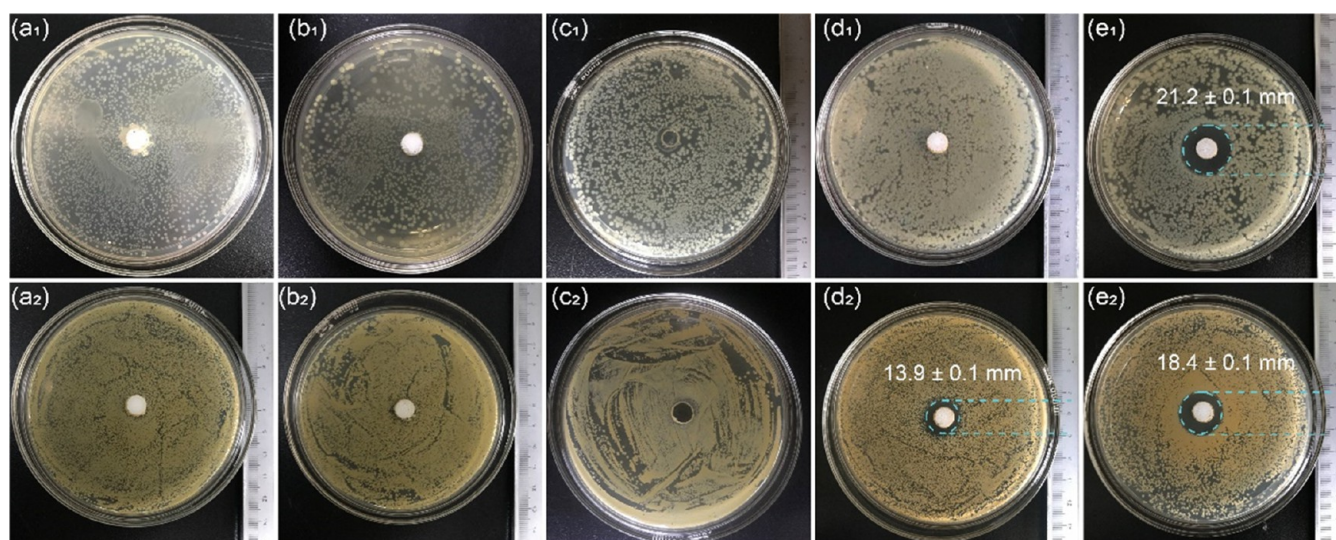


Figure 4. Digital photographs of inhibition zones against *E. coli* (first row) and *S. aureus* (second row) growth after being cultured for 24 h at 37 °C with (a₁, a₂) pristine h-BN, (b₁, b₂) BNNS, (c₁, c₂) free ADM, (d₁, d₂) BNNS-IL, and (e₁, e₂) BNNS-IL-ADM. Living bacterial count for *E. coli* and *S. aureus*: 1×10^6 CFU·mL⁻¹.

observed that the soil with BNNS-IL-ADM almost totally settled down in the ethanol suspension with a clear solution, indicating that the soil suspension with BNNS-IL-ADM also preferred to be stable in aqueous solutions. After centrifugation, the solid sample was completely removed to obtain clear ethanol solvents, whereas raw soil is difficult to collect via centrifugation owing to its inherent hydrophilicity (Figure S4h). In the meantime, the supernatants in the ethanol aqueous solution presented a more turbid milky white color than in the case of aqueous solution, indicating that an ethanol–water solvent is more conducive to recover BNNS-IL-ADM from soil segments, in line with the more natural color of soil to solve the residual problem (Figure S4h–j).

3.6. Antibacterial Activity In Vitro. In the above-mentioned context, the nanodelivery system of BNNS-IL-ADM could control the sustained release behavior of the cargo (i.e., antibacterial ingredient, ADM). Such aspects are then factored in the performance of antibacterial activity. The disc punching method was applied to further qualitatively compare the inhibitory effect of BNNS-IL-ADM with free ADM (equivalent Mo content) against Gram-negative *E. coli* and Gram-positive *S. aureus*. The digital photographs of the antibacterial effect via incubation for 24 h are displayed in Figure 4 using the diameter of inhibition zones as criteria. BNNS-IL-ADM exhibits a stronger and excellent inhibitory effect against Gram-negative *E. coli* with larger inhibition zones than against Gram-positive *S. aureus* (Figure 4e₁–e₂). Unfortunately, free ADM with an equivalent Mo content cannot obviously inhibit the bacteria regardless of it being Gram-negative and Gram-positive (Figure 4c₁–c₂), which may be probably because the ADM at the loose-dose level is easy to leach in agar without effective direct contact with the bacteria. The comparison of two such scenarios strongly supports the idea that BN-IL-ADM as a nanocarrier facilitates and enhances antibacterial performance via controlling the sustained release behavior of the cargo, which in turn reduces the usage and improves the efficiency of the cargo.

In addition, the pristine h-BN and BNNS possess no inhibitory zone against two such bacteria (Figure 4a₁,a₂,b₁,b₂), indicating the antibacterial activity stemming from the IL and

ADM. Meanwhile, BNNS-IL only holds a certain inhibitory zone against Gram-positive *S. aureus*, while it is not obvious against Gram-negative *E. coli* (Figure 4d₁–d₂). Comprehensively, the inhibitory activity of BNNS-IL-ADM on Gram-negative bacteria is mainly owing to Mo ion clusters preferring to penetrate the bacterial cell membrane where peptidoglycan and an outer lipopolysaccharide layer coexist, whereas Gram-positive bacteria are considered to be less sensitive to Mo ion clusters than Gram-negative bacteria, due to the complete peptidoglycan layer within the cell wall (i.e., *S. aureus*),^{8,15} but imidazolium cations and Mo ion clusters within BNNS-IL-ADM provide a synergistic inhibitory effect on Gram-positive bacteria.

Following the qualitative investigation, the minimal inhibitory concentration (MIC) and minimal bactericidal concentration (MBC) against *E. coli* and *S. aureus* were obtained to quantitatively analyze the antibacterial activity, as depicted in Figure 5a–f. Compared with free ADM, BNNS-IL-ADM possesses a much lower MIC against *E. coli* and *S. aureus* with the corresponding value of 0.16 and 0.24 mg·mL⁻¹ and also a lower MBC of 0.20 and 0.28 mg·mL⁻¹, respectively (Figure 5a₁–a₅,c₁–c₅). In contrast, the free ADM has no bacteriostatic activity against *E. coli* in the concentration range of 0.08–0.24 mg·mL⁻¹ and *S. aureus* in the range of 0.20–2.00 mg·mL⁻¹, until their concentration increased to 2.00 and 3.00 mg·mL⁻¹ against *E. coli* and *S. aureus*, respectively. The corresponding antibacterial rate was evaluated and gradually increased with increasing ADM concentration, Figure 5e,f. When the concentration of ADM is 0.12 mg·mL⁻¹, the antibacterial rate of ADM and BNNS-IL-ADM against *E. coli* is 19.17 and 82.00%, respectively. A further increase in the concentration to 0.16 mg·mL⁻¹ leads to the increase in the antibacterial rate of BNNS-IL-ADM against *E. coli* to 100%, whereas this value of ADM reaches 100% at 2.00 mg·mL⁻¹. Likewise, the antibacterial rate of BNNS-IL-ADM against *S. aureus* also exhibits a similar trend with the case against *E. coli*. Thus, the MIC values of ADM against *E. coli* and *S. aureus* are determined to be 2.00 and 3.00 mg·mL⁻¹, respectively (Figure 5b₁–b₇,d₁–d₇). It can be elucidated that BNNS-IL-ADM improves the antibacterial efficiency by 92.00% in view of the

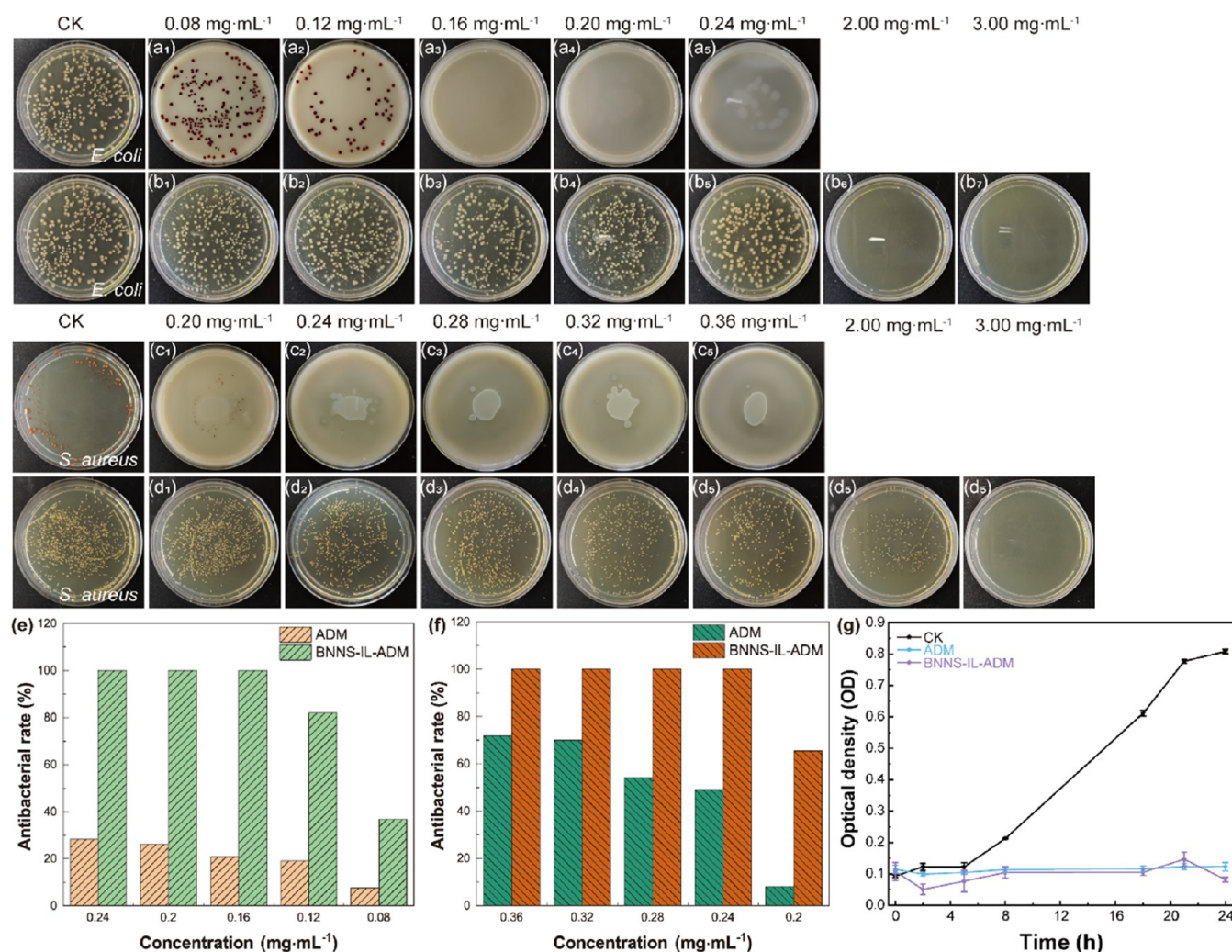


Figure 5. Minimal inhibitory concentration of free ADM and BNNS-IL-ADM against *E. coli* and *S. aureus* based on counting the colonies after culturing for 24 h at 37 °C, where CK was a blank control only with agar and *E. coli* or *S. aureus* in the plate, (a₁–a₅) BNNS-IL-ADM, (b₁–b₇) free ADM, (c₁–c₅) BNNS-IL-ADM, and (d₁–d₇) free ADM under different concentrations; (e) antibacterial rate against *E. coli* with free ADM and BNNS-IL-ADM in the concentration range of 0.08–0.24 mg·mL⁻¹, (f) antibacterial rate against *S. aureus* with free ADM and BNNS-IL-ADM in the concentration range of 0.20–0.36 mg·mL⁻¹, (g) OD growth curves against *E. coli* in LB broth at 37 °C after the bacterial cell was treated with free ADM and BNNS-IL-ADM. The growth of bacteria without any treatment as a blank control (CK), where the concentrations of free ADM and BNNS-IL-ADM were 4 and 0.48 mg·mL⁻¹, respectively, while the living bacterial count for *E. coli* was 1×10^6 CFU·mL⁻¹.

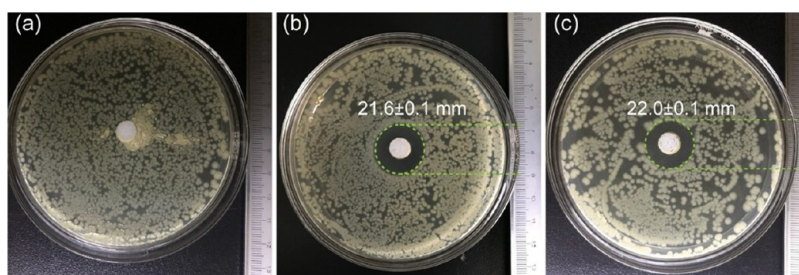


Figure 6. Digital photographs of inhibition zones against *E. coli* growth after being cultured for 24 h at 37 °C with BNNS-IL-ADM with different contents of ADM addition: (a) BNNS-IL-ADM-0.25, (b) BNNS-IL-ADM-0.50, and (c) BNNS-IL-ADM-1.00. Living bacterial count for *E. coli*: 1×10^6 CFU·mL⁻¹.

MIC value against *E. coli* and *S. aureus* in comparison with ADM, where the nanoconfinement effect within BNNS-IL-ADM plays a crucial role. Such features indicate that the introduction of BNNS-IL-ADM enhances the antibacterial activity of ADM at a lower concentration range, which may be presumably ascribed to ADM anchored on the BNNS-IL

lamellar structure augmenting the contacting area with bacteria. As controls, BNNS and BNNS-IL possess much lower antibacterial rates than in the case of BNNS-IL-ADM in the same concentration ranges as depicted in Figure S10.

Additionally, the growth curves of bacteria and bacterial inhibition activity after treatment with free ADM and BNNS-

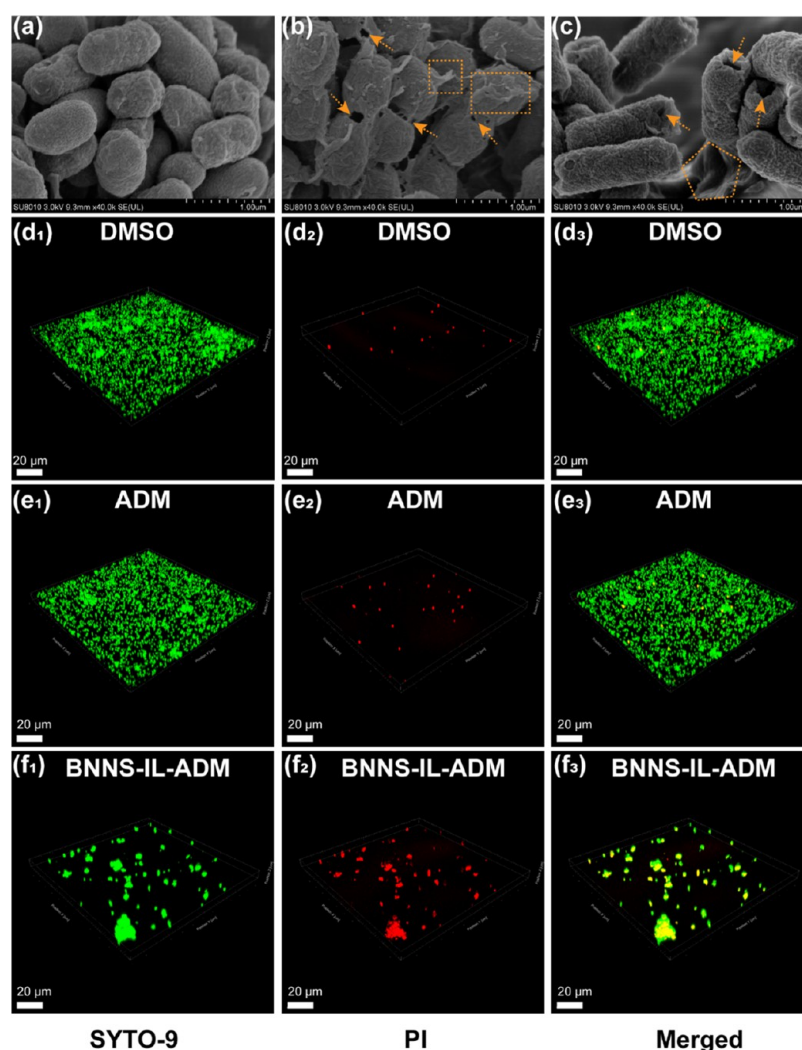


Figure 7. SEM observation images of (a) bare *E. coli* cells and (b, c) cells after coculturing with BNNS-IL-ADM after 6 h captured from different regions (orange arrows indicating the damaged regions and orange boxes representing sample adhesion); confocal laser scanning microscopy images of *E. coli* cells adhered on samples after 12 h (green and red fluorescence representing the SYTO-9-stained living cells and the PI stained dead cells, respectively, and merged indicating the merging of green and red fluorescence), (d_{1–3}) DMSO-treated bacteria, (e_{1–3}) free ADM-treated bacteria, and (f_{1–3}) BNNS-IL-ADM-treated bacteria.

IL-ADM were characterized by detecting the optical density (OD) values of *E. coli* in the suspension via a microplate reader with the change of culture time, as described in Figure 5g. The results show that the OD value of the blank control (CK) grows sharply with the prolonging of continuous culture time, and a higher OD value means a larger concentration of bacteria.⁷⁴ Conversely, both free ADM and BNNS-IL-ADM exhibit inhibition of growth periods with negligible OD value growth within 24 h, and much lower OD values indicate excellent bacterial inhibition behavior. It can also be elucidated that BNNS-IL-ADM enhances the antibacterial activity of ADM with a lower concentration than free ADM.

Moreover, the effects of the ADM content within BNNS-IL-ADM (labeled as BNNS-IL-ADM-0.25, -0.50, and -1.00) on the antibacterial activity against *E. coli* using the disc punching method were investigated as shown in Figure 6. The equivalent dosage of BNNS-IL-ADM with different ratios of ADM (0.25, 0.50, 1.00) produces differentiated inhibition zone diameters. In detail, the low content of ADM (i.e., ratio of 0.25) shows no obvious inhibition effect against *E. coli* (Figure 6a); nevertheless, the inhibition zone is obviously expansionary with the

increase of the ADM content within BNNS-IL-ADM (Figure 6b,c). Such a phenomenon demonstrates that the antibacterial activity mainly stems from Mo ion clusters of ADM and exhibits positive correlation dependence on its concentration.

3.7. Antibacterial Synergistic Mechanism. To further understand the interaction between BNNS-IL-ADM and bacterial cells, biological SEM was used to observe the change of the bacterial surface morphology after treatment with BNNS-IL-ADM. As a demonstration, Gram-negative bacteria *E. coli* was selected as the representative pathogenic microorganism. In comparison with bare *E. coli* cells retaining the integrated cellular structure with a rod-like shape and blunt round ends in Figure 7a, once treated and cocultured with BNNS-IL-ADM, BNNS-IL-ADM nanosheets reveal strong adhesion and attachment to the membrane of bacterial cells by hydrophobic interactions, electrostatic interactions, van der Waals forces, etc., exhibiting a more rough surface and disrupted holes, and the direct contact with the bacterial suspension in turn leads to physical rupture or damage of the cell membrane and destroys the integrity of the bacterial cell membrane due to nanosheet-like sharp edges, as shown in

Scheme 2. Schematic Diagram of the Synergistic Mechanism of BNNS-IL-ADM against Bacteria

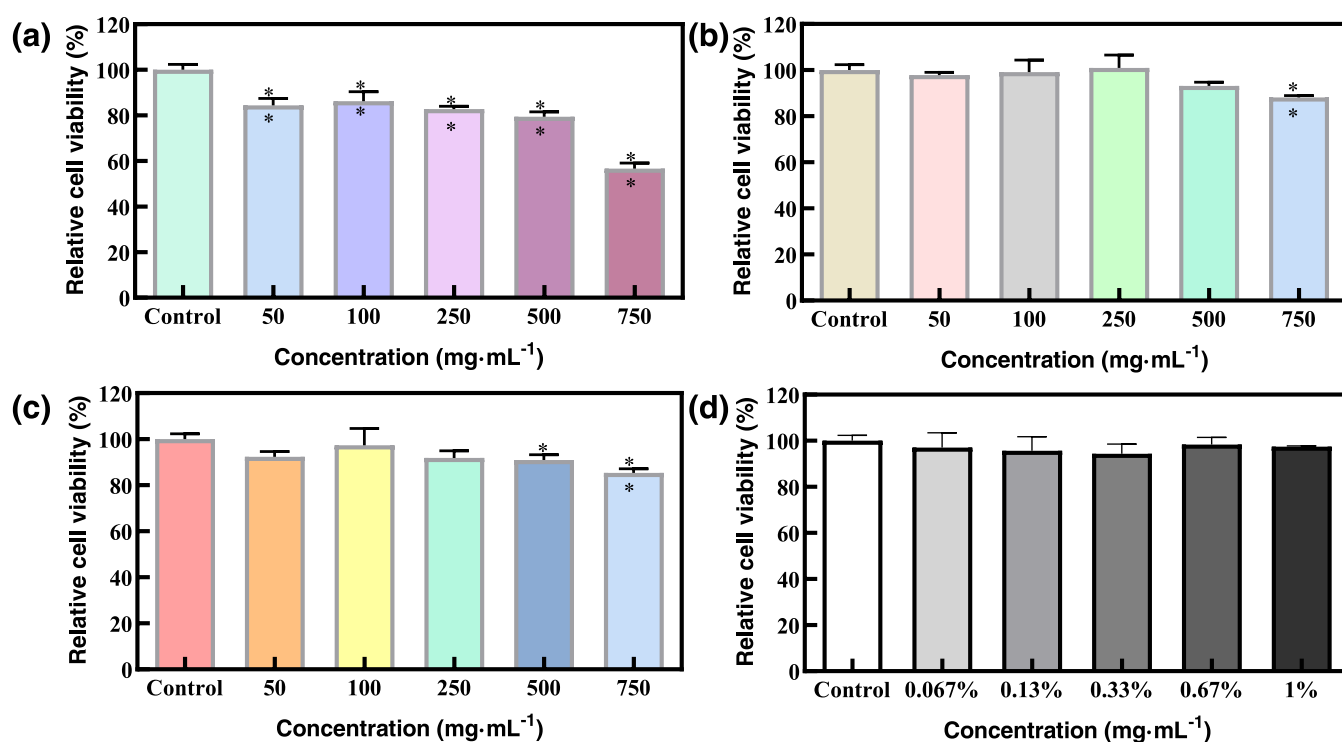
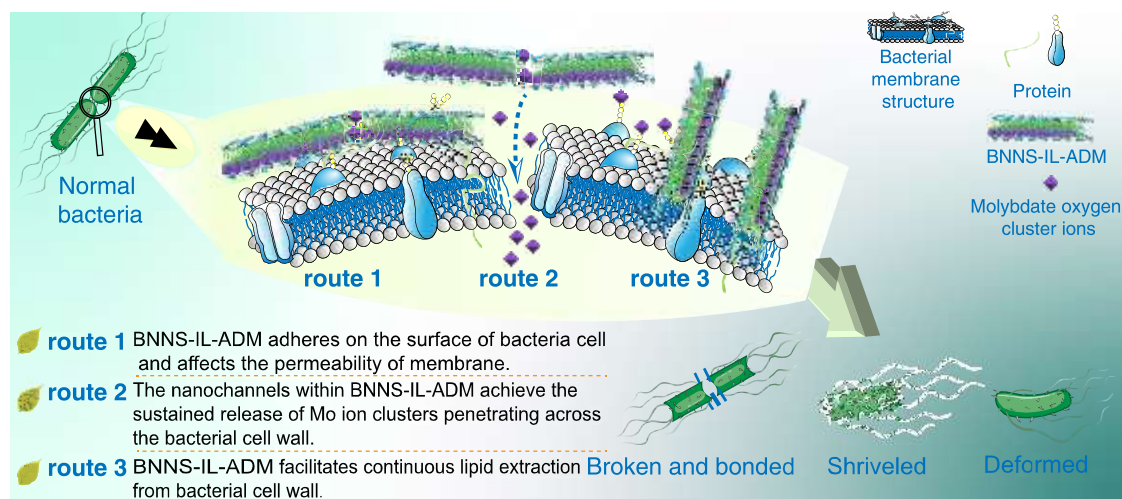


Figure 8. Cell viability of (a) BNNS, (b) free ADM, (c) BNNS-IL-ADM, and (d) DMSO on NIH3T3 cells.

Figure 7b,c. Furthermore, the interaction between the inherently hydrophobic BNNS and cell membrane promotes continuous extraction of the lipid from the bacterial cell wall,⁷⁵ which can have a deleterious impact on the bacterial cell membrane damage. After direct contact, BNNS-IL-ADM can deliver Mo ion clusters across the bacterial membrane and cause deformation in the membrane shape, Figure 7b,c. The apoptotic states of the cell membrane further induce severe problems related to the intracellular component leakage along with cell integral fragmentation,⁴ which confirmed the synergistic antibacterial effect of BNNS-IL-ADM.

Additionally, images of cell fluorescence staining were also collected to investigate the behavior of killing bacteria. The representative model *E. coli* with a concentration of 1×10^6 CFU·mL⁻¹ was stained with SYTO-9 and PI and then detected

by confocal laser scanning microscopy after 24 h of treatment, as illustrated in Figure 7d–f. As reported, SYTO-9 can selectively penetrate the cell membrane of living bacteria once immersed into the cell matrix and stain the nucleic acid of normal growth cells by displaying green fluorescence at 503 nm, whereas PI only stains the membrane structure of damaged or dead cells, showing red fluorescence at 620 nm.^{15,76} After treating and coculturing with 0.01 wt % DMSO as a control as well as free ADM with a concentration of 0.48 mg·mL⁻¹, most of the *E. coli* cells in the 3D platform emit green fluorescence in Figure 7d_{1–3},e_{1–3}, indicating free ADM without good inhibition ability at low concentrations below the MIC value. Instead, after coculturing with BNNS-IL-ADM for 24 h, the dead or damaged bacterial cells exhibit a sharp growth with the red fluorescence region spreading in Figure

$7f_{1-3}$. The aggregation and cluster state of cells in the 3D images also demonstrated the enhanced adhesion ability between cells in the presence of BNNS-IL-ADM, which exacerbated the destruction of the cell membrane and apoptosis, thus affecting the functions of *E. coli* in the aspects of nutrient uptake and respiratory exchange.

In this context, BNNS-IL-ADM with the potent capability of nanoconfinement serves the purpose of a delivery carrier for the inherent antibacterial agent ADM, exhibiting excellent local action without loss, which exerts a superior antibacterial effect especially at a low concentration level of bioactive components. The mechanism of the synergistic effect may be summarized as follows (as depicted in Scheme 2): (i) the cell adhesion at the initial phase is presumably ascribed to the inherent hydrophobic property of the BNNS as well as electrostatic interactions and van der Waals forces, (ii) BNNS-IL-ADM facilitates continuous lipid extraction from the bacterial cell wall,³⁰ and (iii) the nanochannels within BNNS-IL-ADM protect the bioactive ingredient and achieve the sustained release of Mo ion clusters penetrating across the bacterial cell wall.

3.8. Cytotoxicity Analysis. Given that BNNS-IL-ADM improves antibacterial efficiency, the cytotoxicity of the nanomaterial is a crucial issue during practical applications. Regarding this, the cytotoxicity toward murine fibroblast cells (NIH3T3) was evaluated in vitro using colorimetric methyl thiazolyl tetrazolium (MTT) assay. The cell viability of NIH3T3 cells within 24 h was monitored to investigate the cell growth situation in the presence of BNNSs, ADM, and BNNS-IL-ADM in the concentration range of 50–750 mg·mL⁻¹, individually. As shown in Figure 8a, the BNNS exhibits a significant concentration-dependent effect on the cell viability, which is attributed to the good adhesive ability of the BNNS on cell walls. Indeed, the BNNS presents low cytotoxicity with a cell viability of above 80% at the concentration of the BNNS less than 250 mg·mL⁻¹. In fact, the concentration during practical applications is usually below 30 mg·mL⁻¹. In addition, ADM possesses no cytotoxicity at low concentration; excitingly, the cell viability is still 88% even when the concentration of ADM reached up to 750 mg·mL⁻¹ (Figure 8b). Intriguingly, the novel strategy of exfoliation and functionalization to facily prepare BNNS-IL-ADM obviously promotes the cell viability in sharp contrast to BNNSs. This tendency shown in Figure 8c indicates that the cytotoxicity of BNNS-IL-ADM is not obviously concentration-dependent, which exhibits negligible cytotoxicity below 500 mg·mL⁻¹ and low cytotoxicity with a cell viability of 85% toward murine fibroblast cells even at high concentrations up to 750 mg·mL⁻¹. As a control, the solvent DMSO aimed at dispersing the BNNS and BNNS-IL-ADM has no cytotoxicity, Figure 8d. Hence, the BNNS-IL-ADM nanomaterial can be considered a safe and efficient 2D delivery platform for biological applications.

4. CONCLUSIONS

In summary, BNNS-IL-ADM is successfully prepared via ionic exchange interactions to promote BNNS exfoliation and conjugation with ADM. The obtained BNNS-IL-ADM effectively forms BNNSs with three to four layers and micron lateral dimensions with a large aspect ratio of ~325. Such planar BNNS-IL with alignment of cations and anions within BN nanochannels provides enough and continuous space for ADM intercalation with a Mo element content of 6.25% and inhibits their aggregation via weakening their attraction force.

The nanochannels within BNNS-IL-ADM provide nanoconfinement inducing capability for sustained release of Mo ions by shifting down near 30.45%, fitting best with the first-order kinetic model. The time of half-release extends to 3.67 h for the case of BNNS-IL-ADM in comparison with 1.37 h for free ADM. BNNS-IL-ADM facilitates and enhances the antibacterial efficiency by 92.00% in view of the MIC value against *E. coli* and *S. aureus* in comparison with ADM. BNNS-IL-ADM with potent capability of inducing a nanoconfinement effect serves the purpose of a delivery carrier for the inherent antibacterial agent ADM, exhibiting excellent local action without loss, which exerts a superior antibacterial effect especially at a low concentration level of bioactive components. The nanoconfinement protective effect and strong bacterial cell adhesive ability stemming from BNNS-IL-ADM played a synergistic role with bioactive ADM molecule sustained release against bacterial infection. BNNS-IL-ADM can serve as an efficient and safe antibacterial agent with very low cytotoxicity.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsanm.3c00247>.

Additional experimental details, including materials, exfoliation, and hydroxylation of h-BN, recyclability of IL, dispersion stability, sustained release behavior experiments, and statistical analysis, EDS mapping, AFM micrograph, and topographical height profiles, colloidal stability and adhesion ability, XRD patterns, Raman spectra, XPS full spectrum, high-resolution XPS spectra of B 1s and N 1s electrons, TG and DTG curves, DSC thermograms, cumulative Mo ion release curve in aqueous medium, minimal inhibitory concentration of BNNS-IL and BNNS against *E. coli* and *S. aureus*, and release kinetic model fitting data of free ADM and BNNS-IL-ADM (PDF)

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Notes

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